

modularity of the digital medium definitely offers an advantage in configuring a work for physical space. It also means that an installation becomes just one possible version of a piece—a version that might never be reinstalled elsewhere. Because new media art is more process-oriented than object-oriented, it is important to convey the underlying concept of this process to the audience.

PLATFORMS OF EXCHANGE

In new media exhibitions, various exchanges occur between the institution, curator, artist(s), and audience and create a highly complex matrix of relationships. New media art requires a close collaboration between the artists and curators and a continuous discussion about the presentation of a work. The role of a new media curator is increasingly less that of “caretaker” of objects (as the original meaning of the word “curator” suggests) and more that of a mediator and interpreter or even producer. A curator often mediates between the artist and the institution, which often must create formats and procedures to accommodate new media art; between the artwork and the general audience unfamiliar with new media art and in need of guidance and explanation; and between the artwork and the press. With a continuing shortage of new media experts at traditional newspapers and art magazines, curators often must furnish detailed explanations of the work.

The development of the work and its presentation in a physical space require close collaboration between curators and artists (sometimes several artists for a single work). The collaborative model is also crucial to the artistic process itself. Besides the often complex collaboration of new media artists, programmers, researchers, and scientists (whose role may range from consultant to full collaborator), some works begin with the artist’s establishing a framework in which other artists create original works. Lisa Jevbratt’s *Mapping the Web Infome* and Alex Galloway and the Radical Software Group’s *Carnivore* are perfect examples.⁷ In each case, artists set parameters by means of software or a server and invite other artists to create “clients,” which in and of themselves again constitute artworks. The initiating artist plays a role similar to that of a curator, and the collaboration often results from extensive discussions (sometimes on mailing lists established for the purpose). Showing these works in a museum context may lead to yet another level of curatorial “intervention.” Collaboration and exchange are also inherent in the broader culture of the networked digital medium and an important element in multiuser

environments—for example, three-dimensional worlds that rely on their inhabitants to extend the world and create dwellings—and gaming.

ENGAGING THE AUDIENCE

An important step in getting new media art out of its ghetto and integrating it into the art world is to broaden its audience. That is feasible only if institutions and curators facilitate exchanges with and about the artwork.

Visitors to an exhibition in a traditional art institution cannot be described accurately as the audience, a label that suggests a unified, homogenized group of people and neglects their diverse backgrounds and social contexts. The online, “virtual” audience for software, Internet, and game art mostly consists of self-organizing communities of interest that are embedded in different “networked cultures,” each with its own emphasis—on art, social systems, activism, programming, gaming, and so forth. Most of those who attend new media festivals are knowledgeable about the field and not especially diverse. Once new media art is introduced into the more traditional museum or gallery space, it is exposed to a more diverse audience that does not consist predominantly of experts and perceives this art form largely as something radically new. They play the most important role in integrating new media art into the museum gallery.

The museum/gallery audience for new media art might be divided roughly into the following categories: the “experts” who are familiar with the art form; the fairly small group of those who claim a “natural” aversion to computers and technology and refuse to look at anything presented by means of them; a relatively young audience segment that is highly familiar with virtual worlds, interfaces, and navigation paradigms but not necessarily accustomed to art that involves these aspects; and those who are open to and interested in the art but need assistance using it and navigating it.

“Getting it right” for all these groups is challenging, if not impossible. New media curators working in an institutional context encounter recurring criticisms—voiced by audience members, art critics, or the institution itself—that usually contain at least a kernel of truth. These complaints, some of which are discussed here, are a helpful “reality check” for the reception of new media art. Moreover, they effectively highlight some of the art’s distinguishing characteristics.

"IT'S ALL ABOUT TECHNOLOGY"

New media art, to a greater or lesser extent, is "about technology." No object or art form (painting, sculpture, or photography) can be separated from its own materiality, and one could argue that every painting also is "about" painting and comments on its own medium—although self-reflexivity substantially varies from one work to another.

In most cases, this complaint about technology expresses frustration with its gratuitous use—showcasing technology for its own sake. Applied to new media art, this critique is linked to a person's familiarity with the medium. Gratuitous use of technology can only produce bad art. Technology is a medium, like paint or clay, for most new media artists. Having worked with it for a decade, if not several decades, they take it for granted. This is not to say that these artists are uninterested in or do not closely follow the "latest" technologies. Because the medium often lags behind the concepts that artists try to communicate, they must often push the boundaries or develop technologies to express their ideas.

If a museum visitor is unfamiliar with a specific technology or interface, it automatically becomes the focus of attention—an effect unintended by the artist. For the expert audience, in contrast, the technology is transparent and thus moves to the background and becomes mostly a vehicle for content. Unfortunately, such variations of focus and perception cannot easily be addressed. Art audiences and museum visitors have looked at paintings for centuries, and for many the medium of paint is neither a surprise nor an obstacle. But the cultural heritage that has "trained" us in approaching certain art forms, such as painting, has not necessarily provided us with a vocabulary to understand others, such as new media.

An additional factor that needs to be considered here is that every emerging medium explores its own characteristics as a necessary and important step in shaping artistic practice. Many of Nam June Paik's works—such as *Magnet TV* and *TV Crown*—investigated the "materiality" of television and video.

Moreover, new media art often critically investigates its underlying technologies and their encoded cultural and commercial agenda, automatically, as a result shifting focus to the medium itself. Not until new media art makes regular appearances in the art world will its technologies be taken for granted rather than understood as a fixation.

"IT DOESN'T WORK"

Describing the reaction to E.A.T.'s famous performance exhibition 9 *Evenings: Theatre and Engineering* in 1966, Billy Klüver remarked, "Critics and public had a field day at the engineers' expense. . . . Anything that was assumed to have gone wrong (whether it actually did or not) was attributed to technical malfunctions."⁸ His comment captures one popular strategy of critics: if you cannot denounce the art—for lack of understanding or arguments—attack the technology.

But in fact complaints about nonworking technological art are all too often justified. Unless a venue specializing in this art form organizes the exhibition, new media art is often shown without sufficient or properly maintained technical support. Consequently, the art is undercut and audiences are frustrated. It can be difficult for an audience to distinguish what has failed, the art or the technology.

Institutions must ensure that new media works are adequately supported, but in new media art, technical malfunctions may simply be a fact of life that has to be accepted. Only consider how often office and home computers crash, and it is clear that technology is not infallible. While the industry strives to make its products more stable, digital technologies are developing at a speed that virtually guarantees continuing bugs and glitches. Rather than blame the art, one probably needs to understand technological shortcomings as integral to its content.

"IT BELONGS IN A SCIENCE MUSEUM"

As established boundaries and categories between an art and a science museum erode, the potential for new media art to find a place and relevance in both institutions might come to seem as a strength rather than a shortcoming. In the digital era, the technologies of representation in art and science converge constantly. Even if they differ in focus, both art and science now have to address issues of communication, representation, and simulation in (three-dimensional) networked spaces; information and data management; issues of interfacing as well as ethical implications of their exploration (particularly in biotechnology and genetic engineering). Science more and more relies on simulation in its use of three-dimensional worlds, virtual reality, and immersive environments. Art is exploring the same environments—often using scientific data—in an attempt to construct realities and ways of communicating.

An information-based, networked society emphasizes relationships between bodies of knowledge and necessitates a collaboration of human minds and networks to establish these relationships. Information networks require an interdisciplinary approach, and artists are constantly playing with, appropriating, and exploring scientific findings and data.

Although collaborations between artists and scientists play a major role in new media art and discourse, these explorations are not necessarily met with enthusiasm by either the scientific community or the art world, which seldom acknowledges them (the 1986 Venice Biennale was devoted to the relationship between art and science). Art and science can benefit tremendously from each other's approaches, however. As art, science has created its own language and metaphors and could profit from artistic projects that explore these aspects of representation. In its many crossovers into other disciplines—among them various sciences—new media art could support a more holistic approach to culture and help us bridge the gap between the “two cultures” of the sciences and the humanities that C. P. Snow famously outlined in 1959.⁹

Art and science have always been closely linked, and their relationship has invariably been a complex and often uneasy one that has shifted and developed, so that art and science have become attached at certain points and have grown apart at others. Both Raphael, in his use of perspective, and Leonardo da Vinci are said to have married art and science. The 2003 showing of Leonardo's *Leicester Codex* in New York both at the Museum of Natural History and, later, at the Metropolitan Museum of Art across Central Park, seems to testify to that marriage. Leonardo seems to have faced some of the criticisms voiced today about new media art. One of his drawings exhibited at the Met included the handwritten note of a “critic” who suggested that Leonardo was neglecting art because of his preoccupation with technology. The digital age has the potential to bridge gaps between art and science and, at least theoretically, bring them closer.

Developments in art and science have always affected our understanding of reality. We attempt to locate and quantify our awareness of both internal processes and external objects, states, or facts through systems and representations. Artistic and scientific technologies of representation both reflect and structure our awareness of the culture we are embedded in. Observing and representing used to be primarily object-oriented—what is represented is seen. Developments in theoretical science (from quantum physics to chaos theory and fuzzy logic), as

well as in digital art (interactive, networked projects, virtual reality, etc.) suggest a shift from the object as a form of truth to conditions of possibility. These developments were to some extent mirrored in the critical theory of poststructuralism and postmodernism.

The spaces between the actual and the virtual worlds and realities, the gaps and overlaps between these different spaces and states, including subjectivity and objectivity, constitute an underlying concern of both art and science. "Virtual reality" (in the broadest sense) is not simply a useful method of simulation; it is a platform for exploring our "being" in different worlds, virtual and actual.

The networking of science, technology, and aesthetics often ends up in a simple visualization of abstract data represented in a diagrammatic structure. What is lacking are insights about the connection between reality and the autonomy of images, since three-dimensional visuals tend to be identified with a representation of "real" objects. Many new media art projects have investigated the question how scientific knowledge may be translated into aesthetics, and whether there are possibilities for new visuals without simple visualization. The achievement of these projects consists in creating a dialogue on the interaction between the actual, the virtual, and the hypothetical—which potentially is of great benefit to both the arts and sciences. Defining precise contexts in which new media art should or should not exist runs counter to both the intrinsic qualities of the art itself and the stage our culture finds itself in today.

"I WORK ON A COMPUTER ALL DAY—
I DON'T WANT TO SEE ART ON IT IN MY FREE TIME"

"Computer art" is embedded in our daily lives more than most other art forms—more than video and photography. This is simultaneously a great asset and a great obstacle. On the one hand, the link between computers and the economic, social, and cultural fabric of our media-saturated lives gives new media art relevance and urgency; on the other hand, new media's potential audience may not want to reflect critically on or engage creatively with the medium that also is a major tool in the work environment.

One would expect that video—in its close connection to television, a delivery mechanism for anything ranging from the daily news to "pure" entertainment and an epitome of consumer culture—should provoke some of the same reactions. It may owe its acceptance as a medium for art to its strong connection to home entertainment (rather than to work)

and the presentation strategies employed for showing it in a gallery. Video art is now presented less frequently on monitors and more often shown as projections or even elaborate installations, automatically shifting the context. Exhibiting new media art in an environment that suggests an office (computers and monitors on desks) may sometimes be the best option but inevitably creates certain reception problems.

"I WANT TO LOOK AT ART, NOT INTERACT WITH IT"

Art audiences around the world have long played the role of "art consumers"—a role accommodated by an excessively consumer-oriented culture. Granted, art movements such as the Situationists, Fluxus, or conceptual art also relied on audience participation but they remained exceptions to the rule. Most visitors to a museum or gallery go with the expectation of seeing a "selection" of high-quality art for contemplation. At the same time, we always interact with art—engaging with it or even "completing" it. This interaction, however, remains a highly personal affair, and traditional art objects require no active, physical engagement to reveal themselves. Art that breaks with the conventions of contemplation and purely private engagement shocks the average museumgoer, disrupting the mind-set that art institutions so carefully cultivated. Most individuals experience their most direct involvement with art and its tools in school; in museums, participatory art-related "activities" are confined mostly to workshops and tours for children and families. In general, "creativity" in art is nurtured primarily in children and young adults.

Ideally, new media artworks themselves should inspire interaction, but given the context of traditional museum culture, institutions may need to take the initiative to overcome the reluctance of the public to engage with the art. Exchange—encouraged and made appealing through docents, instructions, and an inviting setup of artwork—has to become part of the curatorial concept.

Interaction alone does not take art to a higher level or constitute quality in and of itself. It is simply a reality of contemporary artistic practice. As the artist David Rokeby puts it, "Interaction is banal. We talk to each other on the street. We breathe in air, modify it chemically, then breathe it back out to be breathed in by others. We drive cars. We make love. We walk through a forest and scare a squirrel. I am looking forward to a time where interaction in art becomes as banal and unremarkable . . . merely another tool in the artistic palette, to be used when appropriate."¹⁰

"WHERE ARE THE SPECIAL EFFECTS?"

The frustration with the perceived gratuitous use of technology in new media art is counterbalanced by the common criticism that new media art does not live up to the visual standards set by digital entertainment. The digital entertainment industry, which has become important to cultural life, has also led to a profound misunderstanding of what new media art might or should be. According to Norman Klein, in his essay "Inside the Stomach of the Dragon: The Victory of the Entertainment Economy," "Terms like consumerism and mass culture seem naive now. We all essentially live inside the stomach of the 'entertainment' dragon. As a result, it would be near impossible to generate an avant-garde strategy in a world that feels increasingly like an outdoor shopping mall, what I call a scripted space."¹¹ Klein refers to our era as that of the "Electronic Baroque," a term Angela Ndalani also uses in the title of her book *Neo-Baroque Aesthetics and Contemporary Entertainment*, in which the neo-baroque is a model for understanding today's films, computer games, and theme park attractions.¹²

The neo-baroque digital entertainment industry, with its ever bigger, better, and more sophisticated special effects, has helped to create a society of the digital spectacle that needs to satisfy its consumers' unending demand for the next level of attractions. New media art, with its link to digital technologies, is often subjected to similar demands. If it does not dazzle with the latest effects, it is considered "lame." Art resides in the realm of sculpture and painting; new media need to entertain.

Even a sophisticated art audience sometimes switches to new criteria in evaluating new media art, measuring the design of an art project's virtual world by the standards of commercial games with million-dollar budgets and a design team of dozens of people. The art project may be a complex and advanced investigation of the navigation paradigms of its commercial counterpart, but the art audience's desire for visual effects is sometimes stronger than its interest in a critical exploration of human-computer interaction or paradigms of agency and control. New media art's proximity to the entertainment industry can also prove highly problematic in its integration into the art world. One might argue that this is ultimately a nonissue, since art institutions—with stores selling coffee mugs, posters, and T-shirts embellished by art—already reside "in the stomach of the dragon."

The criticisms I have discussed here offer a glimpse of the relatively "unsafe" place new media occupy in relation to the art world at large.

It is essential that both curators and institutions be aware of new media's precarious position and open up spaces where this position can be discussed. This means a diversified approach to platforms of exchange and "interface" with the audience. These attempts at interfacing must not, however, become overly didactic, making art only a vehicle for educating the public.

The digital medium, with its flexibility and amenability to customization, allows a more active involvement by the audience in the curatorial process. One can find quite a few examples in the online art world of essentially "self-organizing" portals and repositories that allow the public to participate in a curatorial process of selecting, evaluating, and featuring artworks. Art institutions, however, neglect the audience, failing to involve it in the curatorial process. The idea of "public curating" currently is in the experimental stage, but there seems to be a growing effort to develop models for such collaboration—both through Web sites and in the gallery space.

In 2001, the Massachusetts Museum of Contemporary Art (MASS MoCA) invited gallery visitors to use a curatorial software program to project their selections from more than one hundred digital images of twentieth-century works of art from the museum's collection onto the walls of the gallery.¹³ The project, called *Your Show Here*, was created by Tara McDowell and Letha Wilson (project coordinators), Chris Pennock (software design), Nina Dinoff (graphic design), and Scott Paterson (information architecture). Visitors could browse through the database of images, filtering works according to artist name, medium, date, and keyword; choose up to five; write a statement about their choices; and title the show. By just clicking a button, visitors could project digital images at the scale of the originals. The virtual exhibition remained in the gallery only until the next participant "installed" new choices, but a printout of each person's curatorial decisions could be posted on the bulletin board near the gallery entrance.

The project used instant recycling, reproduction, and archiving, all facilitated by the digital medium, to propose an alternative model for presenting and viewing art that moves away from more traditional approaches. The art can take on new meanings in multiple reconfigurations. While this model of "public curating" still begins with a specific archive of images, it blurs the boundaries between the public and the curator, allowing an exhibition model that might more directly reflect the demands, tastes, and approaches of an audience. Some will resist the reconfiguration of roles—curator, artist, audience, and museum—brought

about by new media. And although the new model may need time to develop fully, it suggests the potential of digital technologies themselves as an open-source model for creating and presenting art.

One of the challenges that digital interfacing poses for museums is to balance the needs of both traditional art objects and process-oriented (new media) art. There have always been and always will be art objects. Today these are supported by a cultural “system” of presentation and preservation that includes museums, galleries, collectors, and conservators. New media art does not threaten these objects. It now has a place in multiple contexts and will continue to have one even if it should be fully integrated into the art world. The intrinsic features of new media art ultimately protect it from being co-opted by the art establishment. Nevertheless, its integration is in museums’ own best interest: new media art constitutes a contemporary artistic practice that institutions cannot afford to ignore. It can also expand the notion of what art is and can be. Picking up where previous art forms—from kinetic to conceptual art—left off, new media art has the potential to broaden and question our understanding of the history of art.

NOTES

1. See Sabine Himmelsbach, “Vom ‘White Cube’ zur ‘Black Box’ und weiter. Strategien und Entwicklungen in der Präsentation von Medienkunst im musealen Rahmen,” in *Digitale Transformationen*, ed. Monika Fleischmann and Ulrike Reinhard (Heidelberg: WHOIS Verlagsgesellschaft, 2004), 171–73.

2. *Data Dynamics*, Whitney Museum of American Art, 2001; <http://artport.whitney.org/exhibitions/past-exhibitions.shtml>.

3. The memory palace is an old mnemonic device and strategy that is based on the connection between physical and mental space. In the second century BCE, the Roman orator Cicero imagined inscribing the themes of a speech on a suite of rooms in a villa, and then delivering that speech by mentally walking from space to space. Also see “The Art of Memory,” <http://cotati.sjsu.edu/spoetry/folder6/ng621.html>.

4. Mark Napier, *P-Soup*, <http://www.potatoland.org/p-soup>; Andy Deck, *Open Studio*, <http://draw.artcontext.net>.

5. See also Camille Utterback, *Untitled 5*, <http://www.camilleutterback.com/untitled5.html>.

6. See also Scott Snibbe, *Screen Series*, <http://www.snibbe.com/scott/screen/index.html>.

7. Lisa Jevbratt, *Mapping the Web Infome*, <http://www.newlangtonarts.org/network/infome>; Alex Galloway and RSG, *Carnivore*, <http://www.rhizome.org/carnivore>.

8. Billy Klüver, in *The New Media Reader*, ed. Noah Wardrip-Fruin and Nick Montfort (Cambridge, MA: MIT Press, 2003), 212.

9. C. P. Snow, *The Two Cultures* (Cambridge: Cambridge University Press, 1998 [1959]).

10. David Rokeby, lecture at "Info Art," Kwanju Biennale, Korea, 1996, <http://www.interlog.com/~rokeby/install.html>.

11. Norman Klein, "Inside the Stomach of the Dragon: The Victory of the Entertainment Economy," http://www.eyebcam.org/reblog/journal/archives/2005/01/inside_the_stomach_of_the_dragon.html (accessed August 8, 2007).

12. Angela Ndalianis, *Neo-Baroque Aesthetics and Contemporary Entertainment* (Cambridge, MA: MIT Press, 2004)

13. <http://www.massmoca.org>.